

CS1112 - Foundations to CompSci I

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Examinations

2 in person

50% each

1 after set theory finished

Foundations of CompSci

Sets, Logic and Math for Computing:
Chapter 7, 2, 3, 4, 8

← Name of the
text book
(optional)

5, 7, 9, 10 next module

* Book used for different view on same material *

Computer Science:

- Study of
 - Computer Systems
 - Computer Software
 - What is? How does it run, how do we write
 - Computation
 - What can be computed, what are the underlying principles
- Model: representation of (a part of) a world used to explain/predict/demonstrate features of the world that is being modeled
- CompSci is all about models

Teaching Aims:

- How to describe collections of objects
- How to express yourself clearly and precisely
- How to understand requirements
- How to understand arguments
- How to analyze what you have done
- Many techniques used within Computer Science

! Practice Regularly!

Note: Slides, Class recordings, problem classes, solutions and practice is all you need for exam.

Lecture 1 - Algorithms

* IBM - Ponder This *
- puzzles - HARD

- Anything a computer does is based on carrying out a sequence of steps
 - A sequence of steps to achieve a task is an "algorithm"
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⊕ Algorithms:

- Time complexity:

Big O notation - how steps vary with input size n - scalability
◦ $O(n)$

- Space complexity

Sometimes ignored
↑ e.g. traffic lights

An algorithm is an ordered, deterministic, executable, terminating set of instructions

Instructions must be clear, deterministic and precise?

CS1112

Sets, logic, etc